

Riya Joshi

Embedded Systems & IoT Firmware Engineer • 1 Years Experience • MERIQ Index: 98% (Mastered)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: PCB Debugging, MQTT, Python, FreeRTOS, ARM Cortex, Embedded Linux

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core PCB Debugging service, reducing latency by 46%.
- Led a squad of 5 engineers delivering scalable microservices processing 48M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 51%.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [PCB Debugging, MQTT, Python, FreeRTOS]

Engineered end-to-end production solution leveraging PCB Debugging and MQTT with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, FreeRTOS, ARM Cortex, Embedded Linux]

Architected high-resilience system integrating Python with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

Carnegie Mellon University — 2021 - 2025 (Score: 8.22 / 10.0)